

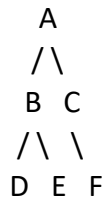
1.

a) What does AI mean?

Artificial Intelligence, here 'Artificial' means 'Man made' and 'Intelligence' means 'Thinking power'. So, AI means Man made thinking power.

b) Prove that uniform-cost search and BFS with constant step costs are optimal when used with the GRPAH-SEARCH algorithm.

let's illustrate these properties with an example. Consider the following graph:

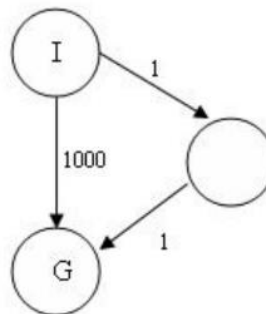


Let's assume that the step cost between each adjacent pair of nodes is 1.

- **Optimality of UCS:** Suppose we want to find the optimal path from node A to node F. UCS will expand nodes in the following order: A, B, C, D, E, F. It selects and expands nodes based on the lowest path cost. In this case, the optimal path is $A \rightarrow C \rightarrow F$ with a total cost of 2.
- **Optimality of BFS with Constant Step Costs:** BFS will explore nodes level by level. In this case, it will expand nodes in the following order: A, B, C, D, E, F. Since all step costs are the same, BFS guarantees that the first time it reaches node F, it has found the shortest path.

c) Show a state space with constant step costs in which GRAPH-SEARCH using iterative deepening finds a suboptimal solution. Compare the four evaluation criteria set of several uninformed search strategies.

The state space that iterative deepening search with GRAPH-SEARCH algorithm will fail to find the optimal solution can be below figure where I is the initial state and G is the goal state.



- **Evaluation criteria set of several uninformed search strategies**

BFS	Yes	yes	$O(b^d)$	$O(b^d)$
DFS	No	No	$O(b^m)$	$O(bm)$
UCS	Yes	yes	$O(b^{1+(C^*/\epsilon)})$	$O(b^{1+(C^*/\epsilon)})$
DLS	Yes	no	$O(b^l)$	$O(bxl)$
IDDFS	Yes	Yes	$O(b^d)$	$O(bd)$
Bidirectional	Yes	Yes	$O(b^{d/2})$	$O(b^{d/2})$
BFS	Yes	yes	$O(b^d)$	$O(b^d)$

- d) What is the Heuristic function of the informed search strategy? How to minimize the total estimated solution cost using the best-first search, A* search algorithm. Show the heuristic must be consistent for the optimal solution in the A* algorithm.

Heuristic function: Estimated cost of the cheapest path from node n to a goal node. $f(n)=g(n)+h(n)$

Minimize the total estimated cost using A* search:

By making A* admissible, we can minimize the total estimated solution cost.

For A* search algorithm the equation we use is

$$F(n)=g(n)+h(n)$$

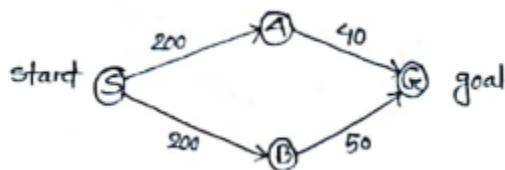
Where $g(n)$ = actual cost from start node to n

$h(n)$ =estimated cost from n to goal node.

We know, when:

$h(n) \geq$ actual cost, then it is overestimation and when

$h(n) \leq$ actual cost then it is underestimation



Here,

$$g(A)=200$$

$$g(B)=200$$

actual cost, $A \rightarrow G=40$

actual cost, $B \rightarrow G=50$

case I: Overestimation

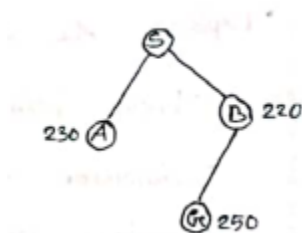
let, $h(A) = 80=280$

$$h(B)=70$$

$$\text{now, } f(A)=200+80=280$$

$$f(B)=200+70=270$$

$$f(G)=g(A)+h(G)=250+0=250$$



case II: Underestimation

let, $h(A) = 30$

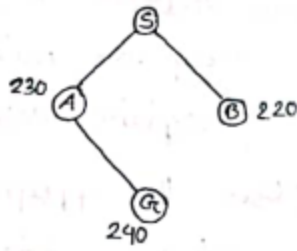
$$h(B) = 20$$

$$f(A)=200+30=230$$

$$f(B)=200+20=220$$

$$f(G)=250$$

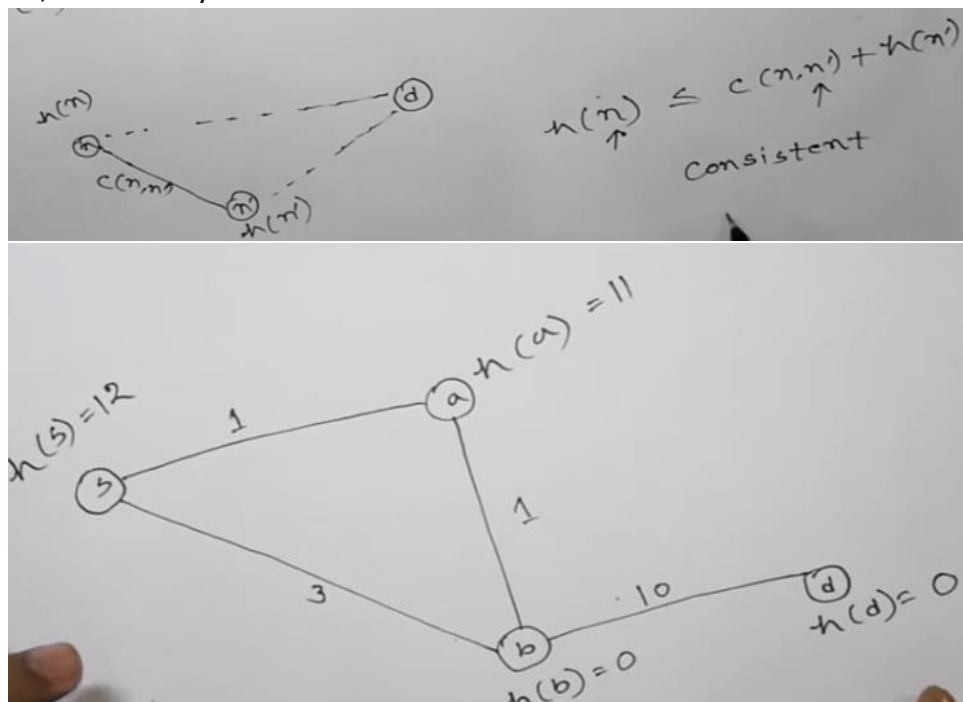
$$f(G)+g(G)+h(G)=200+0=240$$



The consistency condition ensures that the heuristic provides a meaningful estimate of the cost to reach the goal, and it plays a crucial role in guaranteeing the optimality of A*.

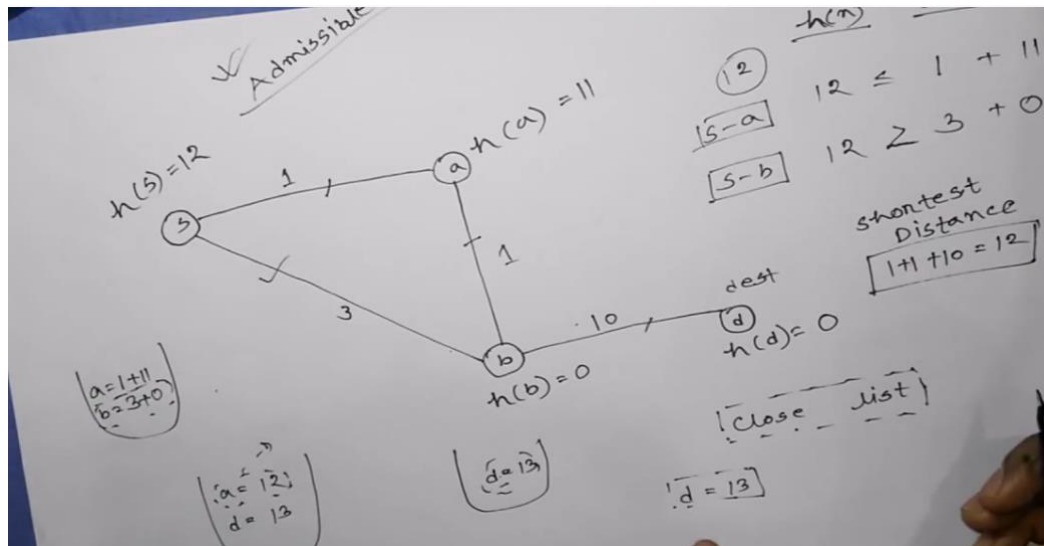
Admissible Heuristic:

- **Definition:** A heuristic that never overestimates the true cost to reach the goal.
- **Illustration of Admissible Heuristic:** $h(n) \leq \text{actual cost}$, if this relation maintains for every node, then we say this Admissible Heuristics.



➤ Consistent heuristics:

- **Definitions:** A heuristic that is admissible and satisfies a consistency condition.
- **Illustrations:** This graph is not consistent hence it does not provide any optimal solution.



2.

a) Why do we use a local search strategy to address the optimization problem? What are the key advantages of local search algorithm?

Local search strategies are used for optimization problems due to their efficiency in exploring large solution spaces, adaptability to different problem types, iterative improvement process, capability to handle non-convex spaces.

➤ **Advantages:**

- Use very little memory.
- Can find reasonable solutions in large or finite state.

b) Show how the last configuration of 4-queens on a 4x4 board has fewer conflicts than the first configuration using a local search strategy in where conflicts mean there are no two queens on the same row, column or diagonal.

Initial state:

	1	2	3	4
1	Q			
2		Q		Q
3				
4	Q			

- Queen at (1,1) conflicts with the queen at (2,2) diagonally.
- Queen at (2,2) conflicts with the queen at (2,4) Horizontally.
- Queen at (1,1) conflicts with the queen at (4,1) vertically.

Final state:

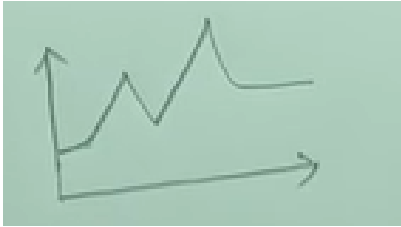
	1	2	3	4
1		Q		
2				Q
3	Q			
4			Q	

- No queen conflict with others.

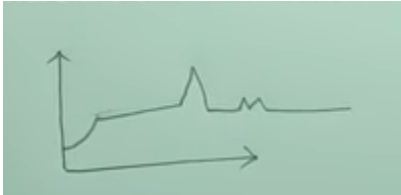
c) What are the reasons and problems of the hill-climbing algorithm for getting stuck? How to escape these problems using the Simulated-annealing search algorithm?

➤ Hill-climbing often gets stuck for the following reasons:

➤ **Local maxima:** Algorithm can't go higher.



➤ **Plateaux:** Area where the evaluation function is flat.

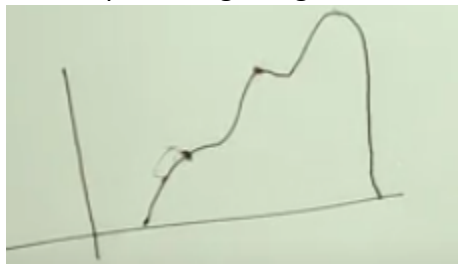


➤ **Ridges:** Search may oscillate slowly.



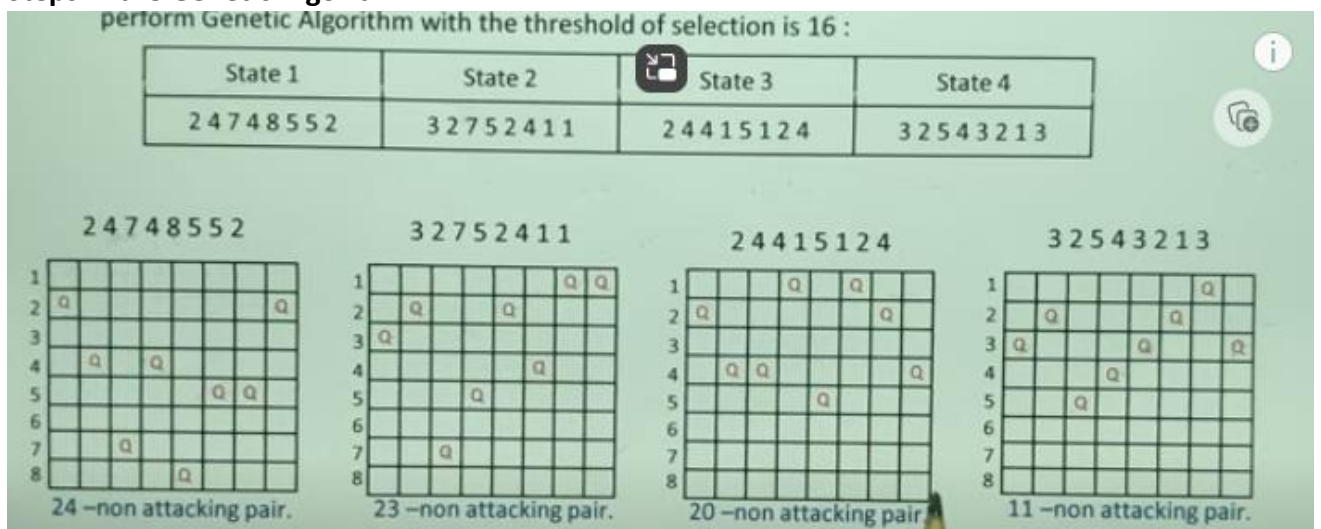
➤ Escape these problems using the Simulated-annealing search algorithm:

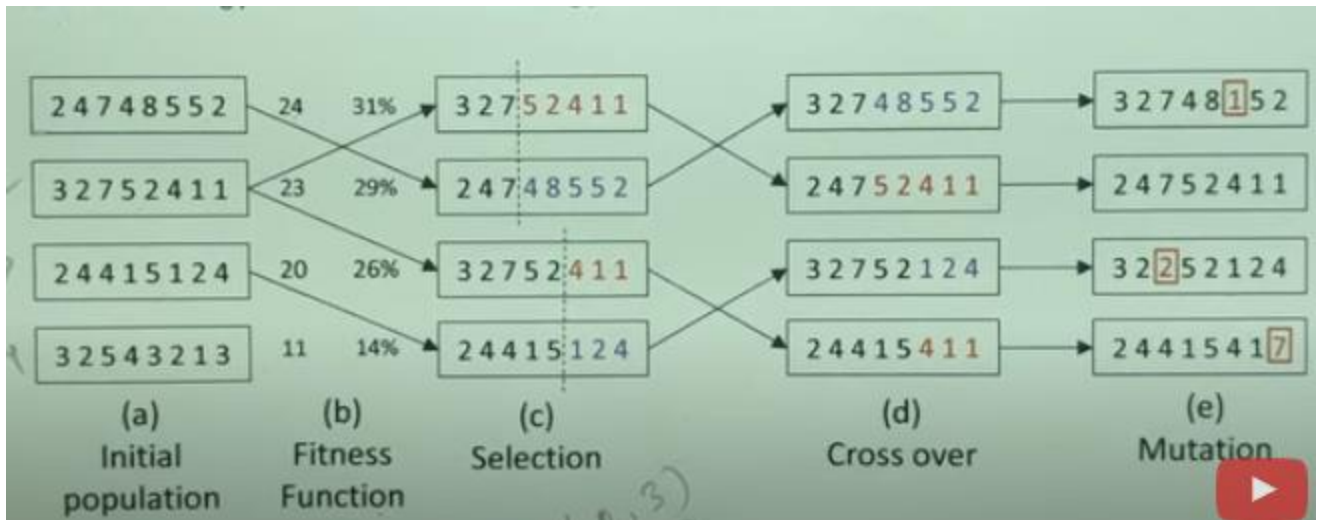
- Hill-Climbing always goes uphill, looking for the best solution nearby. Sometimes, it might get stuck on a hill and miss a better solution on a different hill. Simulated Annealing is like Hill-Climbing, but with a trick. It sometimes goes downhill a bit, allowing it to explore other hills. This helps avoid getting stuck and find a better overall solution.



d) Illustrate and explain the genetic algorithm using digit strings representation of 8-queen states.

Steps in the Genetic Algorithm:





3.

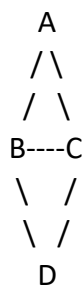
- Define the following terms in your own words Conditional probability, Marginal probability, Normalization and independent probability.
- What is Bayes rule? Computer the patient's probability of having liver disease if they are an alcoholic. "Being an alcoholic" is the test (kind of like litmus test) for liver disease. Past data tells you that 10% of patients entering your clinic have liver disease and 5% of the clinic's patients are alcoholics. You might also know that among those patients diagnosed with liver disease. 7% are alcoholics.
- Design a naïve Bayes model, a Bayesian classifier based on the dentistry example.

4.

- What is supervised learning? How to learn decision trees using entropy and information gain of attributes.
- What is univariate linear regression? How to minimize the loss using gradient descent for fitting linear regression?
- Define constraint satisfaction problems. Represent the map coloring problem with a constraint graph.**

A problem is solved when each variable has a value that satisfies all the constraints on that variable is called CSP.

Map coloring: Let's consider a simple example with four regions: A, B, C, and D. The constraint graph can be represented as follows:



In this graph:

- ✓ Node A is connected to nodes B and C.
- ✓ Node B is connected to nodes A and C.
- ✓ Node C is connected to nodes A, B, and D.
- ✓ Node D is connected to node C.

Each node is associated with a set of possible colors, typically represented as $\{R, G, B\}$ for red, green, and blue. The goal is to find a coloring assignment that satisfies the constraints (no two adjacent nodes have the same color).

A possible initial state could be:

A: $\{R, G, B\}$

B: $\{R, G, B\}$

C: $\{R, G, B\}$

D: $\{R, G, B\}$

Then, as the constraint satisfaction algorithm progresses, certain colors might be eliminated based on the neighboring nodes' colors until a valid coloring is found.

d) Define robot. Briefly describe different types of robot hardware.

A **robot** is a programmable machine designed to perform tasks autonomously or semi-autonomously, typically in the physical world.

Robot hardware:

1. **Actuators:** Motors, servo motors.
2. **Sensors:** Vision sensors, infrared sensors, ultrasonic sensors, touch sensors.
3. **Power Supply:** Batteries.
4. **Processors/Controllers:** Microcontrollers, CPUs.
5. **Communication Modules:** Wireless modules, sensor interfaces.
6. **Frames/Chassis:** Structural components, wheels, tracks, legs.

5.

a) Consider the problem of deciding whether a propositional logic sentence is true in a given model.

- I. Write a recursive algorithm $PL-TRUE?(s, m)$ that returns true if and only if the sentence s is true in the model m (where m assigns a truth value for every symbol in s). The algorithm should run in time linear in the size of the sentence. (Alternatively, use a version of this function from the online code repository.)

```
function PL-TRUE?(s, m) returns true or false
  if s = True then return true
  else if s = False then return false
  else if SYMBOL?(s) then return LOOKUP(s, m)
  else branch on the operator of s
    ¬: return not PL-TRUE?(ARG1(s), m)
    ∨: return PL-TRUE?(ARG1(s), m) or PL-TRUE?(ARG2(s), m)
    ∧: return PL-TRUE?(ARG1(s), m) and PL-TRUE?(ARG2(s), m)
    ⇒: (not PL-TRUE?(ARG1(s), m)) or PL-TRUE?(ARG2(s), m)
    ⇔: PL-TRUE?(ARG1(s), m) iff PL-TRUE?(ARG2(s), m)
```

Figure S7.2 Pseudocode for evaluating the truth of a sentence wrt a model.

- II. Give three examples of sentences that can be determined to be true or false in a partial model that does not specify a truth value for some of the symbols.

b. The question is somewhat ambiguous: we can interpret "in a partial model" to mean in all such models or some such models. For the former interpretation, the sentences $False \wedge P$, $True \vee \neg P$,

and $P \wedge \neg P$ can all be determined to be true or false in any partial model. For the latter interpretation, we can in addition have sentences such as $A \wedge P$ which is false in the partial model $\{A=\text{false}\}$.

c. A general algorithm for partial models must handle the empty partial model, with no assignments. In that case, the algorithm must determine validity and unsatisfiability,

III. Show that the truth value (if any) of a sentence in a partial model cannot be determined efficiently in general.

IV. Modify your PL-TRUE? algorithm so that it can sometimes judge truth from partial models, while retaining its recursive structure and linear run time. Give three examples of sentences whose truth in a partial model is not detected by your algorithm.

It helps if and or evaluate their arguments in sequence, terminating on false or true arguments, respectively. In that case, the algorithm already has the desired properties: in the partial model where P is true and Q is unknown, $P \vee Q$ returns true, and $\neg P \wedge Q$ returns false. But the truth values of $Q \vee \neg Q$, $Q \vee \text{True}$, and $Q \wedge \neg Q$ are not detected.

b) Which of the following is correct?

In all cases, the question can be resolved easily by referring to the definition of entailment.

- a. $\text{False} \models \text{True}$ is true because *False* has no models and hence entails every sentence AND because *True* is true in all models and hence is entailed by every sentence.
- b. $\text{True} \models \text{False}$ is false.
- c. $(A \wedge B) \models (A \Leftrightarrow B)$ is true because the left-hand side has exactly one model that is one of the two models of the right-hand side.
- d. $A \Leftrightarrow B \models A \vee B$ is false because one of the models of $A \Leftrightarrow B$ has both A and B false, which does not satisfy $A \vee B$.
- e. $A \Leftrightarrow B \models \neg A \vee B$ is true because the RHS is $A \Rightarrow B$, one of the conjuncts in the definition of $A \Leftrightarrow B$.
- f. $(A \wedge B) \Rightarrow C \models (A \Rightarrow C) \vee (B \Rightarrow C)$ is true because the RHS is false only when both disjuncts are false, i.e., when A and B are true and C is false, in which case the LHS is also false. This may seem counterintuitive, and would not hold if $\Rightarrow$ is interpreted as “causes.”
- g. $(C \vee (\neg A \wedge \neg B)) \equiv ((A \Rightarrow C) \wedge (B \Rightarrow C))$ is true; proof by truth table enumeration, or by application of distributivity (Fig 7.11).
- h. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$ is true; removing a conjunct only allows more models.
- i. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B) \wedge (\neg D \vee E)$ is false; removing a disjunct allows fewer models.
- j. $(A \vee B) \wedge \neg(A \Rightarrow B)$ is satisfiable; model has A and $\neg B$.
- k. $(A \Leftrightarrow B) \wedge (\neg A \vee B)$ is satisfiable; RHS is entailed by LHS so models are those of $A \Leftrightarrow B$.
- l. $(A \Leftrightarrow B) \Leftrightarrow C$ does have the same number of models as $(A \Leftrightarrow B)$; half the models of $(A \Leftrightarrow B)$ satisfy $(A \Leftrightarrow B) \Leftrightarrow C$, as do half the non-models, and there are the same numbers of models and non-models.

- c) Consider a vocabulary with the following symbols:(8.10)
 Occupation (p, o): Predicate. Person p has occupation o.
 Customer (p1,p2): Predicate. Person p1 is a customer of person p2.
 Boss (p1,p2): Predicate. Person p1 is a boss of person p2.
 Doctor , Surgeon, Lawyer , Actor : Constants denoting occupations.
 Emily, Joe: Constants denoting people.

Use these symbols to write the following assertions in first-order logic:

- a. Emily is either a surgeon or a lawyer.
 - b. Joe is an actor, but he also holds another job.
 - c. All surgeons are doctors.
 - d. Joe does not have a lawyer (i.e., is not a customer of any lawyer).
 - e. Emily has a boss who is a lawyer.
 - f. There exists a lawyer all of whose customers are doctors.
 - g. Every surgeon has a lawyer.
- d) What is a language model? Describe the N-gram character models.

6.

- a)
 - b)
- I. **Every gardener likes the sun.**
 a. $\forall x \text{ gardener}(x) \rightarrow \text{likes}(x, \text{Sun})$
 - II. **You can fool some of the people all of the time.**
 a. $\exists x \forall t \text{ person}(x) \wedge \text{time}(t) \rightarrow \text{can-fool}(x, t)$
 - III. **You can fool all of the people some of the time.**
 a. $\forall x \exists t (\text{person}(x) \rightarrow \text{time}(t) \wedge \text{can-fool}(x, t))$
 b. $\forall x (\text{person}(x) \rightarrow \exists t (\text{time}(t) \wedge \text{can-fool}(x, t)))$
 - IV. **All purple mushrooms are poisonous.**
 a. $\forall x (\text{mushroom}(x) \wedge \text{purple}(x)) \rightarrow \text{poisonous}(x)$
 - V. **No purple mushroom is poisonous.**
 a. $\neg \exists x \text{ purple}(x) \wedge \text{mushroom}(x) \wedge \text{poisonous}(x)$
 b. $\forall x (\text{mushroom}(x) \wedge \text{purple}(x)) \rightarrow \neg \text{poisonous}(x)$
 - VI. **There are exactly two purple mushrooms.**
 a. $\exists x \exists y \text{ mushroom}(x) \wedge \text{purple}(x) \wedge \text{mushroom}(y) \wedge \text{purple}(y) \wedge \neg(x=y) \wedge \forall z$
 $(\text{mushroom}(z) \wedge \text{purple}(z)) \rightarrow ((x=z) \vee (y=z))$
 - VII. **Clinton is not tall.**
 a. $\neg \text{tall}(\text{Clinton})$
 - VIII. **All men are not fool.**
- c) What is neural network? Describe the multilayer perception neural network.

1.

a) What are the learning outcomes of AI?

1. **Problem Solving:** AI enables effective problem-solving by analyzing data and generating solutions.
2. **Natural Language Processing:** Enables machines to understand and respond to human language.
3. **Machine Learning:** Allows systems to improve performance through experience and data.
4. **Data Analysis:** Processes vast amounts of data quickly, extracting meaningful insights.
5. **Decision Support:** Provides information to assist human decision-making processes
6. **Robotics:** Empowers machines to perform physical tasks, enhancing productivity.

b) Define in your own words the following terms: **agent, agent program, rationality, autonomy, deterministic and stochastic.**

1. **Agent:** An agent is anything that can be viewed as perceiving its environment through sensors and environment acting upon that environment through actuators.
Example: A self-driving car navigating through traffic.
2. **Agent Program:** The set of rules or algorithms governing the agent's behavior and decision-making.
Example: The algorithms guiding a chess-playing computer program.
3. **Rationality:** The ability of an agent to choose actions that maximize the likelihood of achieving its goals.
Example: A smart thermostat adjusting temperature settings to save energy while maintaining comfort.
4. **Autonomy:** The degree to which an agent can operate independently without direct human intervention.
Example: A drone surveying agricultural fields without constant human control.
5. **Deterministic:** A system or process where the outcome is entirely predictable from its initial state and inputs.
Example: A simple calculator performing arithmetic operations.
6. **Stochastic:** Involving randomness or probability, where outcomes are not entirely predictable and may vary.
Example: Weather forecasting, where predictions involve probabilities due to complex atmospheric systems.

c) What is PEAS in specifying the task environment? Illustrate and describe the structure of the model-based reflex agent.

PEAS is a framework used in artificial intelligence to clearly define the parameters and objectives of a given task or problem, providing a structured way to specify an intelligent agent's characteristics and behavior within a particular environment.

PEAS stands for:

- **Performance measure:** This defines how the success of the agent in the task environment will be assessed.
- **Environment:** This describes the external context or the task environment in which the agent operates.

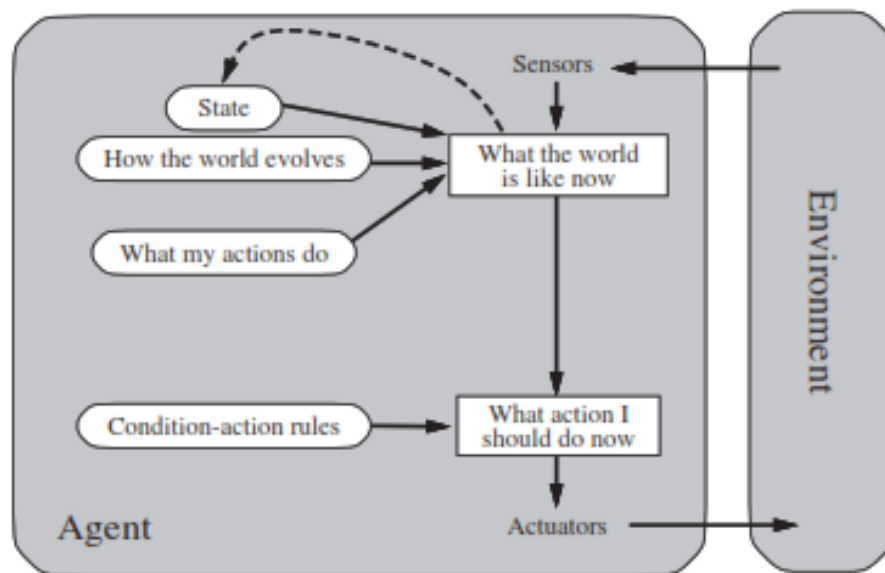
- **Actuators:** These are the mechanisms or components through which the agent affects the environment by taking actions.
- **Sensors:** These are the components that allow the agent to perceive or receive information about the environment.

Example: PEAS description of the task environment for an automated taxi.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

➤ Model-based reflex agents:



2.

- What is uninformed search? Show that the 8-puzzle states are divided into two disjoint sets, such that no state in one set can be transformed into a state in the other set by any number of moves.
- Prove that uniform-cost search and BFS with constant step costs are optimal when used with the GRPAH-SEARCH algorithm.**

Already noted.

- Show a state space with constant step costs in which GRAPH-SEARCH using iterative deepening finds a suboptimal solution. Compare the four evaluation criteria set of several uninformed search strategies.**

Already noted.

3.

- a) What is the Heuristic function of the informed search strategy? How to minimize the total estimated solution cost using the best-first search, A* search algorithm.

Already noted.

Session: 2015-2016.

1.

- a) Define AI within 4 different paradigms with examples.

1. **Thinking Humanly:** The exciting new effort to make computers think.

Example: Chatbots that engage in natural language conversation.

2. **Thinking Rationally:** The study of the computations that make it possible to perceive, reason and act.

Example: Expert systems that use a knowledge base and rules to provide solutions or make decisions in specific domains, like medical diagnosis.

3. **Acting Humanly:** AI strives to mimic human behavior and actions.

Example: Social robots designed to interact with people.

4. **Acting Rationally:** AI focuses on making optimal decisions to achieve specific goals, regardless of human-like reasoning.

Example: Game-playing AI, such as Deep Blue for chess or AlphaGo for Go.

- b) What is rationality? Differentiate between human agent and AI agent in terms of rationality.

- **Rationality:** The ability of an agent to choose actions that maximize the likelihood of achieving its goals.

Example: A smart thermostat adjusting temperature settings to save energy while maintaining comfort.

- **Human Agent:-**

Rationality: Limited by cognitive biases, emotions, and subjective interpretations of information.

- **AI Agent:-**

Rationality: Strives for optimal decision-making based on programmed algorithms, devoid of emotional or biased influences.

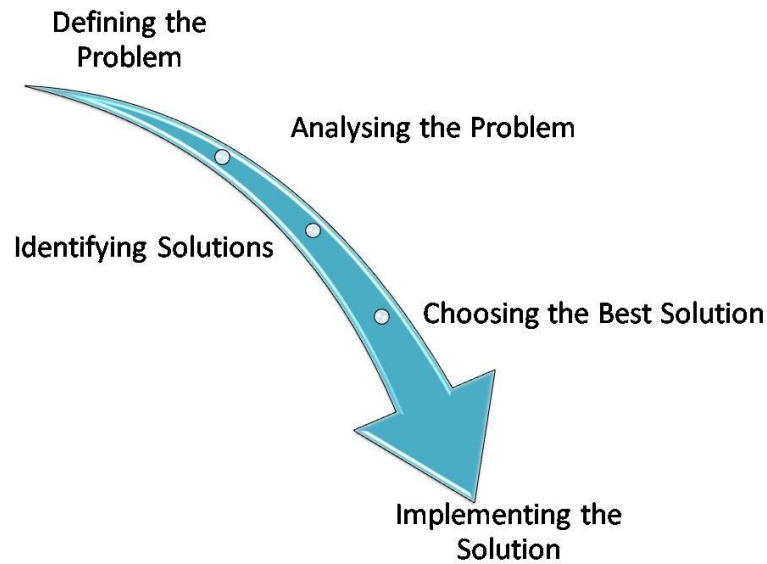
- c) "Do not measure the performance of an agent based on its behavior, Measure the performance of an agent based on what is wanted in the environment."-Justify these statements.

Imagine we have a robot that's supposed to clean a room. If we only look at how the robot moves around (its behavior), we might think it's doing a good job. But what really matters is whether the room gets clean, right?

So, the idea is, don't just watch what the robot does; check if it's actually achieving the goal, which is having a clean room. The quote is saying we should judge the robot based on what we want (a clean room) instead of just looking at how it behaves.

In simpler terms, it's like saying, "Don't judge a cleaning robot by how it moves, judge it by how clean the room gets." This way, we focus on the real purpose, not just the actions.

d) Explain the steps that a problem-solving agent uses to solve a problem with an example.

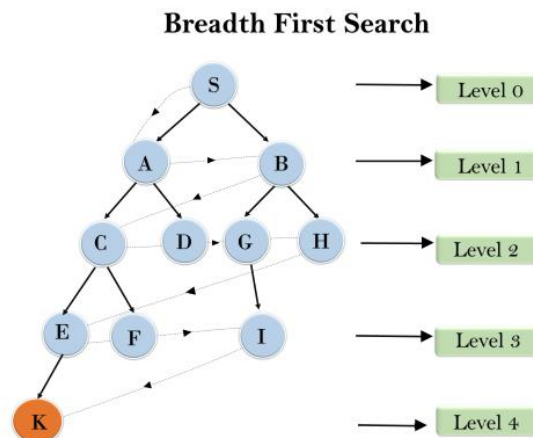


2.

a) **Analyze BFS technique in terms of performance measure. Provide reasons to every measure.**

It visits all the neighbors of a vertex before moving to the next level neighbors.

For example, we have shown the traversing of the tree using BFS algorithm from the root node S to goal node K.



BFS search algorithm traverse in layers, so it will follow the path which is shown by the dotted arrow, and the traversed path will be:

S---> A--->B--->C--->D--->G--->H--->E--->F--->I--->K

➤ **Performance measure:**

- **Completeness:** Yes, BFS is complete if the branching factor is finite.
- **Optimality:** Yes, BFS is optimal if the cost is a non-decreasing function of the depth.
- **Time complexity:** $O(b^d)$ where b is the branching factor and d is the depth of the shallowest solution.
- **Space complexity:** $O(b^d)$ for frontier nodes

b) “Exponential time or space complexity is very expensive”-Justify with an example in terms of time and memory requirements.

Exponential time or space complexity is considered expensive because it grows rapidly with the size of the input, making algorithms impractical for larger instances. The Towers of Hanoi and depth-first search examples illustrate the significant impact on both time and memory requirements when dealing with exponential complexities.

➤ **Justification for Exponential Time and space Complexity:**

Example: The Towers of Hanoi

Explanation: The Towers of Hanoi problem involves moving a tower of discs from one peg to another, with the restriction that only one disc can be moved at a time, and a larger disc cannot be placed on top of a smaller one. The minimum number of moves required is $2^n - 1$ is the number of discs.

Justification: The time complexity and space complexity of a naive recursive solution is exponential $O(2^n)$ as each move leads to two recursive calls. This becomes impractical for large numbers of discs due to the rapid growth in computation time.

c) Compare IDDFS and DFS in terms of performance measure. Which one is preferred when the search space is large and the depth of the solution is unknown? Explain with an example.

Performance of DFS:

- **Completeness:** DFS is not complete if the search space has infinite depth or if there are loops in the graph, as it can get stuck in an infinite loop.
- **Optimality:** DFS does not guarantee optimality; it may find a non-optimal solution if the depth of the goal node is not the shallowest in the search tree.
- **Time complexity:** $O(b^m)$, where b is the branching factor and m is the maximum depth of the search tree. In the worst case, DFS may explore all nodes.
- **Space complexity:** The space complexity of Depth First Search (DFS) is $O(bm)$ in the worst case, where b is the branching factor, and m is the maximum depth of the search tree

Performance of IDDFS:

- **Completeness:** IDDFS is complete as it will eventually find a solution if one exists, exploring deeper levels in each iteration.
- **Optimality:** IDDFS is optimal when the path cost is a non-decreasing function of the depth because it explores shallower levels first.
- **Time complexity:** $O(b^d)$, where b is the branching factor and d is the depth of the shallowest goal node.

Space complexity: $O(bd)$, where b is the branching factor and d is the maximum depth reached in any iteration

Preferred for Large Search Spaces with Unknown Depth:

IDDFS Preferred when the depth of the solution is unknown and the search space is large.

Explanation: IDDFS combines the benefits of DFS with a systematic and controlled exploration of increasing depths. It gradually increases the depth limit, providing a balance between depth-first efficiency and breadth-first completeness.

Example: 8-Puzzle Problem:

Scenario: Solving an 8-puzzle where the optimal solution depth is unknown.

Preference for IDDFS: IDDFS would be preferred due to its ability to perform a series of depth-first searches with increasing depth limits. It avoids the drawbacks of deep exploration while maintaining completeness. This is crucial when the optimal depth is uncertain, and an exhaustive DFS may be impractical.

- d) Consider the following scenario for 8-puzzle game. Figure 3.d.1 is the initial state and figure 3.d.2 is the goal state. Which of the uninformed search or technique would be best suited? Justify answer in terms of performance measure.

3.

- a) How does greedy best first search work? Show using figure-1 what are its properties.
- b) Demonstrate how greedy best-first search is incomplete using figure 1
- c) Demonstrate the working principles of A* search to find a path from Oradea to Bucharest.
- d) Define: Admissible heuristic, Optimally efficient.